

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	YES	Code follows consistent formatting standards.
2	Proper File Structure	Files and folders are logically organized into templates, static, models, data, notebooks, and documentation.	YES	Project structure improves maintainability and navigation.
3	Meaningful Variable Names	Variables are named clearly according to their purpose.	YES	Improves readability and understanding of the code.
4	Function / Method Names	Functions are descriptively named	YES	Naming conventions follow Python best practices.
5	Code Comments	Inline and block comments explain log	PARTIAL	Key sections are documented for easier maintenance.
6	Modular Design	Code is split into reusable functions Functions/modules	YES	Authentication, prediction, database, and utility logic are separated logically.

7	No Redundant Code	Duplicate or unused code is removed	YES	Functions are reused wherever applicable.
8	Error Handling	Exceptions and errors are handled gracefully	PARTIAL	Form validation and application error handling are implemented.

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Flask, Werkzeug	User Registration & Login	High
2	Loan Prediction Module	Python, Scikit-learn	Loan Eligibility Prediction	High
3	Database Module	Flask-SQLAlchemy, SQLite	User Management & Prediction History	High
4	Dashboard Module	Flask, HTML, CSS	Admin, Officer & Analyst Dashboards	Medium
5	Data Preprocessing Module	Pandas, NumPy	Model Training & Prediction	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder hierarchy.
Readability	5	Consistent naming conventions, formatting, and comments improve code readability.
Reusability	5	Core functionalities such as authentication, prediction, and database operations are modular and reusable.
Documentation / Comments	4	Most important modules are documented with comments and supporting documentation.
Overall Score	4.8/5	The project demonstrates clean coding practices, modular architecture, and maintainable implementation.